



Piano Mastery *Portal*

AI-Powered Piano Practice Framework

Upload any piano score — PDF or image — and receive a complete technical analysis, targeted Hanon exercise mapping, a personalized 30-day practice framework, in-app score viewer, floating metronome, split-screen practice mode, and performance video search. All in one beautifully designed portal.

<https://piano-composition-mastery.manus.space>

github.com/BrandonRose2/piano-composition-mastery

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SECTION 01 — 02

Getting Started

01 What Is the Piano Mastery Portal?

The Piano Mastery Portal is a full-stack web application that transforms how pianists approach learning new repertoire. Rather than spending hours searching for practice strategies, the portal uses AI to analyze any uploaded piano score and instantly generate a structured, personalized study plan.

The portal is built around a simple but powerful idea: every piece of music has specific technical demands, and those demands map directly to known exercises. By identifying which Hanon exercises address each challenge in your score, the AI creates a 30-day framework that builds the precise technique you need — not generic drills.

Universal Score Support

Upload any piano score as a PDF or image (PNG, JPG, WEBP). Works for any composer, any era, any difficulty level.

AI-Powered Analysis

GPT-5 Mini reads your score, identifies technical challenges, and generates a complete structured analysis in seconds.

Hanon Exercise Mapping

Each technical challenge is mapped to specific Hanon exercises with rationale, target BPM, and YouTube tutorial links.

30-Day Practice Framework

A personalized daily schedule with milestones, week themes, and measurable goals — all tracked in the app.

LIVE APPLICATION

Access the portal at: <https://piano-composition-mastery.manus.space>

02 Signing In & Your Private Library

The portal uses secure OAuth authentication. Each user has a completely private, isolated library — compositions, progress, and settings are never visible to other users. This makes the portal safe to share with students, colleagues, or the public.

1 Visit the Portal

Navigate to piano-composition-mastery.manus.space. If you are not signed in, you will see the landing page with a sign-in button.

2 Sign In with Manus OAuth

Click **Sign In**. You are redirected to the Manus login portal; after authenticating you return to the home page automatically.

3 Your Library Awaits

Your name appears in the top-right corner. All compositions are stored privately under your account. Sign out at any time via the navigation bar.

Privacy: All data is scoped to your user ID at the database level. Even administrators cannot access another user's compositions or progress records.

SECTION 03

03 Uploading a Score & AI Analysis

The upload workflow is the heart of the portal. Once you provide a score, the AI analysis engine extracts the musical content and generates a complete study package — typically within 15 to 30 seconds.

1 Drag & Drop or Click to Browse

On the home page, drag your score file directly onto the upload zone, or click the zone to open a file browser. Accepted formats: PDF, PNG, JPG, WEBP.

2 File Is Uploaded to Secure Storage

Your score is uploaded to Manus secure cloud storage. For PDFs, the server extracts the text content using pdftotext. The original file is preserved for the in-app score viewer.

3 AI Analysis Begins

The AI (GPT-5 Mini) receives the extracted text and filename, then identifies the composition, composer, key, time signature, technical challenges, and relevant Hanon exercises. A structured JSON response is stored in the database.

4 30-Day Framework Generated

Simultaneously, the AI creates a personalized 30-day practice framework with four weekly themes, daily goals, and three milestone checkpoints.

5 Composition Card Appears in Library

A new card appears in your library with the composition title, composer, difficulty badge, and a progress bar. Click the card to open the full detail page.

SUPPORTED SCORE TYPES

The AI works best with clean, text-searchable PDFs (e.g., IMSLP downloads). Scanned images and handwritten scores are also supported — the AI uses its knowledge of the repertoire to supplement any gaps in text extraction.

🎵 What the AI Produces

Each analysis returns a five-part structured document covering: (1) a historical and musical context overview, (2) a technical challenge breakdown with difficulty ratings, (3) a Hanon exercise mapping table, (4) a 30-day practice framework, and (5) a set of performance principles and practice philosophy notes.

SECTION 04

04 AI Technical Analysis & Hanon Mapping

The detail page for each composition is organized into clearly labeled sections, accessible via the sidebar navigation. The analysis begins with historical context and moves progressively through technical content.

🎵 Historical Context

The first section provides a concise overview of the composition’s historical background, the composer’s intentions, and the musical style. This context helps pianists understand the expressive goals before diving into technique.

🎵 Technical Challenges

The AI identifies the most demanding technical elements of the piece — such as rapid octave passages, hand independence, large interval leaps, or complex rhythmic patterns — and presents each as a labeled challenge card with a difficulty rating and a targeted practice strategy.

🎵 Hanon Exercise Mapping Table

This is the analytical core of the portal. Each identified technical challenge is mapped to one or more specific Hanon exercises from *The Virtuoso Pianist*. The table includes the exercise number, the technical skill it develops, the recommended practice BPM range, and a direct YouTube search button.

HANON #	EXERCISE FOCUS	TARGET BPM	ADDRESSES
1 – 5	Finger independence, evenness	60 → 108	Passage work, scale runs
6 – 10	Thumb crossing, extended positions	60 → 100	Arpeggios, wide stretches
31 – 43	Scales in all major & minor keys	60 → 120	Scale passages, key fluency
44 – 51	Arpeggios (broken chords)	50 → 100	Arpeggio figurations
52 – 60	Trills, tremolos, repeated notes	60 → 132	Trill passages, tremolo

YOUTUBE INTEGRATION

Every row in the Hanon table includes a YouTube button that opens a pre-filled YouTube search for that specific exercise number. No manual searching required.

🎵 Performance Principles

The final section of the analysis presents the AI’s recommended practice philosophy for this specific piece — including guidance on tempo progression, mental practice, section isolation strategies, and performance preparation milestones.

SECTION 05

05 30-Day Practice Framework & Day Tracker

The 30-day framework is not a generic schedule — it is generated specifically for your composition, based on the technical challenges identified in the analysis. The framework is divided into four thematic weeks, each building on the last.

Week 1 — Foundation

Days 1 – 7

Slow, hands-separate practice. Identify problem passages. Establish correct fingering and posture. Begin Hanon warm-up routine at 60% target BPM.

Week 2 — Technique Building

Days 8 – 14

Hands together at slow tempo. Drill the three most technically demanding passages. Increase Hanon BPM by 10% from Week 1 baseline.

Week 3 — Integration

Days 15 – 21

Full run-throughs at 75% tempo. Focus on musical expression and dynamics. Maintain Hanon routine. Begin memorization of key sections.

Week 4 — Performance Preparation

Days 22 – 30

Gradual tempo increase toward performance speed. Record yourself and review. Practice mental performance. Final polish and confidence building.

🎵 Interactive Day Tracker

Each day in the framework is represented by a numbered cell in the tracker grid. Click any cell to expand it and view the day's specific goals, then check it off when complete. Completed days are highlighted in gold. Progress is automatically saved to the database — your tracker state persists across sessions and devices.

Start-Date Picker

Set a start date for Day 1. Every day cell then shows its corresponding calendar date, and the tracker header displays your Day 1 and Day 30 dates.

Milestone Nodes

Three milestone markers (Days 7, 14, and 21) light up in gold when you reach them, providing a sense of accomplishment and momentum.

Progress persistence: For uploaded compositions, progress is stored in the database and syncs across all your devices. For the built-in La Campanella study, progress is stored in localStorage on your current browser.

06 In-App Score Viewer

For any composition uploaded as a PDF or image, the portal renders the score directly in the browser — no need to switch to an external PDF viewer. The score viewer appears as the first section on the detail page, immediately accessible for reference during practice.

PDF Page Navigation

Multi-page PDFs are rendered page by page using pdfjs-dist. Navigate with Previous / Next buttons or jump directly to any page number.

Zoom Control

Zoom from 60% to 300% using the toolbar slider. The score redraws at the new resolution for sharp, readable notation at any size.

Fullscreen Mode

Click the fullscreen button to expand the score to fill the entire browser window. Press Escape or click the exit button to return.

Download Original

A download button in the viewer toolbar lets you save the original uploaded file to your device at any time.

IMAGE SCORE SUPPORT

PNG, JPG, and WEBP scores are displayed with zoom, rotation (90° increments), and fullscreen support. The same viewer handles both PDF and image formats seamlessly.

07 Split-Screen Practice Mode

Split-screen mode is the portal's most powerful practice feature. It places your score and your 30-day tracker side by side in a single view, so you can follow the score while checking off practice tasks — without switching between sections.

1 Activate Split Screen

Click the **Split Screen** button in the composition header (only visible for compositions with an uploaded score file).

2 Drag the Divider

The score appears on the left and the tracker on the right. Drag the central divider handle to resize the panels to your preference.

3 Practice & Track Simultaneously

Navigate score pages, zoom in on passages, and check off completed practice tasks — all in the same view. A fixed top bar shows your overall progress percentage.

4 Exit Split Screen

Click **Exit Split** in the top bar to return to the standard single-column layout.

SECTION 08

08 Floating Metronome Widget

Every composition detail page includes a built-in digital metronome, accessible at any time via the floating button in the bottom-right corner of the screen. The metronome uses the Web Audio API for precise, low-latency beat generation — no plugins or external apps required.

J METRONOME — ALWAYS AVAILABLE

The gold “Metronome” pill button floats in the bottom-right corner of every composition detail page. Click it to expand the full metronome panel; click “Close” to collapse it. The metronome continues running even while you scroll or interact with other sections of the page.

J Metronome Features

FEATURE	DESCRIPTION
BPM Range	40 – 208 BPM, adjustable via slider or ± buttons in 1 BPM increments
Tap Tempo	Tap the “Tap” button repeatedly to set BPM by feel — averages the last 4 taps
Time Signatures	4/4, 3/4, 2/4, 6/8 — the first beat of each bar is accented with a higher-pitched click
Beat Indicator	Visual gold dots light up in sequence with each beat, with the accent beat shown larger
Tempo Reference Marks	Quick-select buttons for standard Italian tempo markings: Larghetto (52), Andante (76), Moderato (108), Allegro (132), Vivace (160), Presto (184)
Quick BPM Presets	One-click buttons for 60, 80, 100, 120 BPM for fast setup

J Using the Metronome for Practice

The recommended approach is to begin each practice session at 60–70% of the target tempo, using the metronome to enforce rhythmic precision. Increase the BPM by 4–8 points only when you can play the passage cleanly three times in a row. The metronome’s tap-tempo feature is particularly useful when you want to match the tempo of a reference recording — simply tap along with the recording to set the BPM.

TECHNICAL NOTE

The metronome uses the Web Audio API’s AudioContext and OscillatorNode for sample-accurate timing. It is not affected by browser tab throttling or JavaScript event loop delays, making it suitable for serious practice sessions.

SECTION 09

09 Performance Video Search

Listening to expert performances is an essential part of learning any piece. The portal searches YouTube for up to **five relevant recordings** of your composition and presents them as selectable cards — so you can pick the correct performance rather than being locked into a single auto-selected result. If none of the search results match, you can paste any YouTube URL directly to override them entirely.

1 Navigate to the Performance Section

On any composition detail page, click **Performance** in the sidebar navigation. The portal automatically searches YouTube using the title, composer, key, and tempo from the AI analysis for a more precise match.

2 Choose from Up to 5 Results

Up to five videos are shown as selectable cards below the main player. Each card displays the thumbnail, title, channel, view count, and duration. Click any card to instantly swap the player to that video. The active card is highlighted with a gold border.

3 Override with a Custom YouTube URL

If none of the search results are correct, click **Paste URL** next to the results header. An input field appears — paste any YouTube URL or bare video ID and press Enter (or click “Use this video”). The player immediately switches to your chosen video. A gold *Custom URL* badge confirms the override is active.

4 Clear the Override

To revert to the search results, click **Clear override** in the metadata card. The player returns to the previously selected search result.

5 Watch Without Leaving the App

Videos play inline within the portal. You can watch a reference performance, then immediately return to the score viewer or tracker — all in the same browser tab.

ACCEPTED URL FORMAT	EXAMPLE
Standard watch URL	youtube.com/watch?v=dQw4w9WgXcQ
Short URL	youtu.be/dQw4w9WgXcQ
Shorts URL	youtube.com/shorts/dQw4w9WgXcQ
Embed URL	youtube.com/embed/dQw4w9WgXcQ
Bare video ID	dQw4w9WgXcQ (11 characters)

Practice tip: Use the performance video alongside the metronome. Watch the performance once for musical inspiration, then set your metronome to 50–60% of the performance tempo and begin your own practice session.

10 Composition Library Management

The home page displays your personal composition library as a grid of cards. Each card shows the composition title, composer, difficulty badge, and a live progress bar indicating how many of the 30 practice days you have completed.

ACTION	HOW TO USE
Open Composition	Click anywhere on the card (outside the title or delete button) to open the full detail page.
Rename Title	Click directly on the composition title text to enter inline edit mode. Press Enter to save, Escape to cancel.
Delete Composition	Hover over the card to reveal the trash icon. Click once to show a confirmation prompt, then click again to confirm deletion.
Progress Bar	Each card shows a gold progress bar indicating completed days (e.g., “12 / 30 days · 40%”). Updated in real time.

BUILT-IN LA CAMPANELLA

The portal always includes a built-in La Campanella (Franz Liszt) study card at the top of the library. This card is available to all users and requires no upload — it uses a pre-authored analysis and framework.

11 Sheet Music Search (IMSLP)

The home page includes a sheet music search bar that queries the IMSLP (International Music Score Library Project) MediaWiki API in real time. IMSLP is the world’s largest free sheet music repository, hosting tens of thousands of public-domain scores.

1 Type a Composition or Composer Name

Enter any search term in the “Find Music & Sheet Music” search bar on the home page. Results appear as you type.

2 Browse IMSLP Results

Each result shows the composition title and a direct link to its IMSLP page, where you can download the free PDF score.

3 Import Directly to Library

Click **Import to Library** on any IMSLP result. A dialog appears where you can paste the direct PDF URL from IMSLP. The portal downloads the file and runs the full AI analysis automatically.

12 Streaming Platform Links

After searching for a composition in the Find Music section, a tab switcher appears below the search bar. Toggle between the **Sheet Music** tab (IMSLP results) and the **Stream** tab to access direct links to the composition on major streaming platforms.

Spotify

Opens a Spotify search for the composition. Works on both the web player and the desktop app.

Apple Music

Opens an Apple Music search. Requires an Apple Music subscription for full playback.

YouTube Music

Opens a YouTube Music search for the composition. Free tier available with ads.

SoundCloud & Tidal

Additional platform cards for SoundCloud (free streaming) and Tidal (high-fidelity audio).

Workflow tip: Use the Stream tab to find a reference recording on Spotify or YouTube Music, listen to it on your phone or second screen, then return to the portal on your main screen for the score viewer and tracker.

13 Import Score from URL

In addition to direct file uploads, the portal can import a score directly from a public URL. This is particularly useful for IMSLP PDFs, which are hosted at predictable public URLs.

1 Find a Direct PDF URL

On IMSLP, right-click the PDF download link and copy the direct URL. It should end in .pdf.

2 Click “Import to Library”

On any IMSLP search result card, click the **Import to Library** button. A dialog box appears with a URL input field.

3 Paste URL and Confirm

Paste the PDF URL and click **Import**. The server downloads the file, stores it in secure cloud storage, and runs the full AI analysis pipeline — identical to a direct upload.

SECTION 14

14 La Campanella — Franz Liszt

The portal ships with a complete, pre-authored study guide for Franz Liszt's *La Campanella* (S. 141/3) — one of the most technically demanding pieces in the standard piano repertoire. This built-in study requires no upload and is available to all users immediately upon signing in.

La Campanella in Brief

Composed 1838, revised 1851. Key of G-sharp minor. Difficulty: **Grade 10 / ABRSM Diploma**. Famous for its bell-like right-hand melody, rapid octave leaps, and extreme hand independence demands. Based on the final movement of Paganini's Violin Concerto No. 2.

🎵 Technical Challenges Covered

- ◆ **Rapid Octave Leaps (RH):** The signature bell melody requires precise, relaxed wrist rotation across large intervals at tempo. Addressed by Hanon exercises 44–51.
- ◆ **Hand Independence:** The left hand maintains a steady accompaniment pattern while the right hand executes the melodic leaps. Addressed by Hanon exercises 1–10.
- ◆ **Chromatic Scale Passages:** Fast chromatic runs in both hands require even finger pressure and thumb-crossing facility. Addressed by Hanon exercises 31–43.
- ◆ **Trill Passages:** Extended trill sections require sustained finger independence without tension. Addressed by Hanon exercises 52–60.
- ◆ **Pedaling & Tone Color:** The bell effect depends on precise una corda and sustain pedal technique to create the characteristic resonance.

🎵 30-Day Framework Highlights

The built-in framework begins with hands-separate practice of the opening theme at 40 BPM, progressing through weekly milestones: the octave leap passage at 60 BPM (Day 7), hands-together at 80 BPM (Day 14), full piece at 100 BPM (Day 21), and a final performance run at target tempo (Day 30).

RECOMMENDED TARGET TEMPO

The standard performance tempo for *La Campanella* is approximately ♩ = 138–152 BPM. Begin practice at 40–50 BPM and increase by no more than 4 BPM per session once a passage is clean.

SECTION 15

15 Feature Quick-Reference Table

The following table provides a complete overview of all portal features, where to find them, and which user actions they support.

FEATURE	LOCATION	KEY ACTIONS
Score Upload	Home — Upload Zone	Drag & drop or click; PDF, PNG, JPG, WEBP
AI Analysis	Composition Detail — Sidebar	Historical context, technical challenges, performance principles
Hanon Mapping Table	Composition Detail — Exercises	View exercise list; click YouTube for tutorial videos
30-Day Tracker	Composition Detail — Practice	Check off days, set start date, view milestones
Score Viewer	Composition Detail — Score	Navigate pages, zoom, fullscreen, download
Split-Screen Mode	Composition Detail — Header	Click Split Screen button; drag divider to resize panels
Floating Metronome	Composition Detail — Bottom-right	Click gold pill to open; set BPM, time sig, tap tempo
Performance Video	Composition Detail — Performance	Up to 5 selectable video cards; click to swap player; Paste URL to override with any YouTube link
Library Cards	Home — Library	Click to open, click title to rename, hover for delete
Progress Bars	Home — Library Cards	Shows completed days / 30 and percentage
IMSLP Search	Home — Find Music	Type to search; click result link or Import to Library
Streaming Links	Home — Find Music (Stream tab)	Click platform card to open search on Spotify, Apple Music, etc.
Import from URL	Home — IMSLP results	Click Import to Library; paste PDF URL; confirm
Start-Date Picker	Composition Detail — Practice header	Click calendar icon; pick date; shows dates on day cells
La Campanella Built-In	Home — Library (first card)	Always available; pre-authored analysis and framework

SECTION 16

16 Tips, Shortcuts & Best Practices

♪ Uploading Scores for Best Results

Text-searchable PDFs (such as those downloaded directly from IMSLP) produce the most accurate AI analyses. If you have a scanned score, the AI will still generate a useful analysis based on the filename and its knowledge of the repertoire, but the technical challenge identification will be less specific. For best results, include the composer's name and composition title in the filename before uploading.

♪ Effective Use of the 30-Day Tracker

The tracker is most effective when used as a commitment device rather than a passive log. Before each practice session, open the portal, read the day's goals, and close the laptop. Practice without distraction. Return to check off the day only after completing the session. This creates a clear ritual boundary between planning and execution.

♪ Metronome Strategy

The “three clean repetitions” rule is the most reliable tempo progression method: increase the BPM only after you can play a passage perfectly three times in a row at the current tempo. Use the tap-tempo feature to match reference recordings exactly, then set a practice tempo at 60–70% of that value.

♪ Split-Screen Workflow

The split-screen mode is most useful during Weeks 2 and 3 of the framework, when you are working hands-together and need to cross-reference specific measures in the score while tracking your daily goals. During Week 1 (hands-separate), the full-page score viewer is typically more useful.

♪ IMSLP Import Workflow

When importing from IMSLP, navigate to the composition's IMSLP page, scroll to the “Sheet Music” section, and click on the PDF file you want. On the IMSLP download page, right-click the download button and select “Copy link address” to get the direct PDF URL. Paste this into the Import dialog in the portal.

GITHUB REPOSITORY

The full source code is available at:

<https://github.com/BrandonRose2/piano-composition-mastery>

Start Your Practice Journey

Visit the portal, sign in, and upload your first score. The AI analysis takes under 30 seconds. Your personalized 30-day framework will be waiting for you.

<https://piano-composition-mastery.manus.space>